

Optoelectronic Performance Modulation in WSe₂/ReS₂ Hetero-transistor via Piezoelectric Stack



Shreyasi Das¹, Srilagna Sahoo¹, Mayank Goyal¹, Abin Varghese² and Saurabh Lodha¹

¹Department of Electrical Engineering, IIT Bombay, India.

²Department of Materials Science & Metallurgy, University of Cambridge, UK.

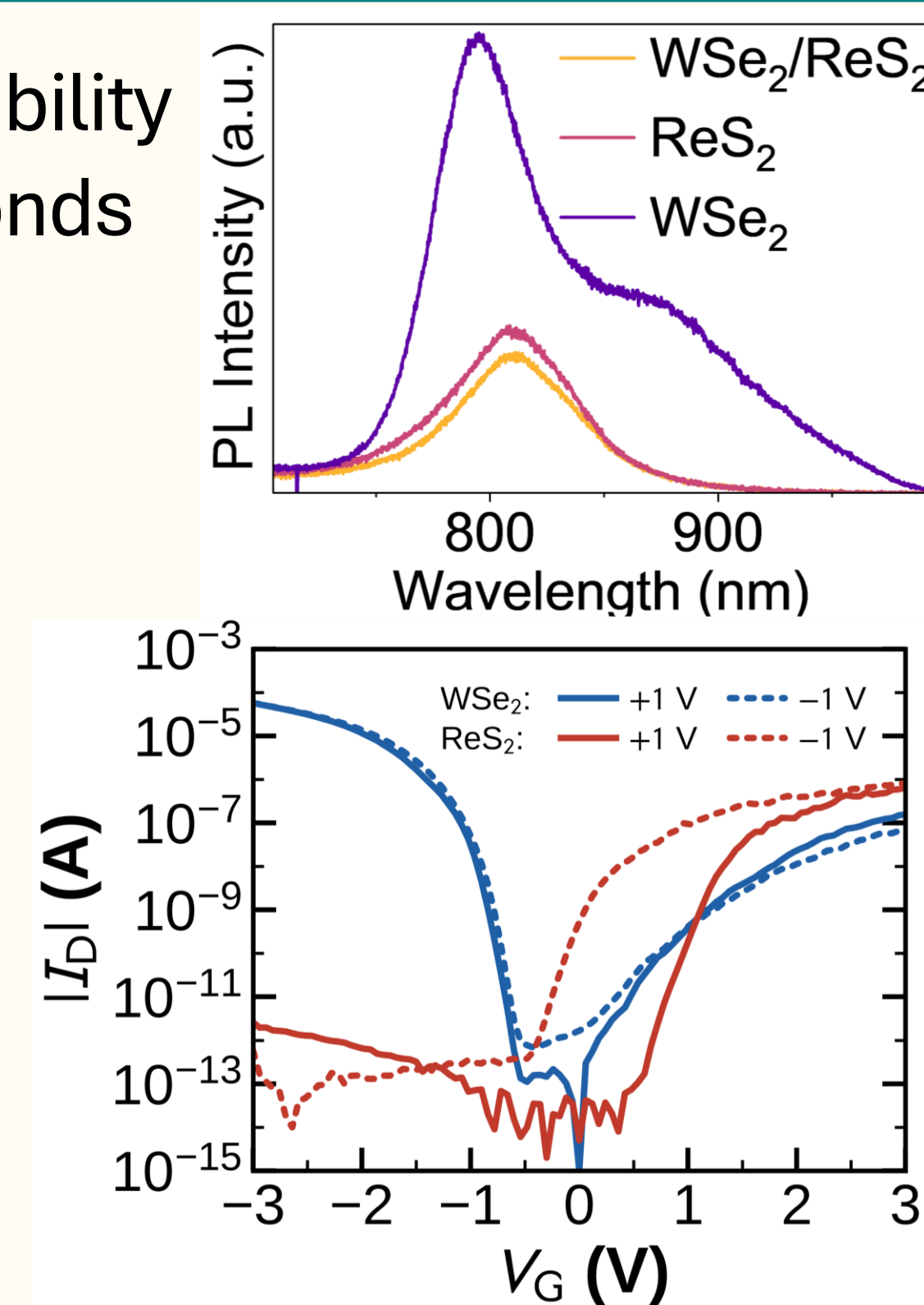


Why 2D TMD for Strain-dependent Optoelectronics?

- Atomically thin materials: ultimate scalability
- Van der Waal's materials: no dangling bonds
- Superior elastic strength
- Contact doping feasible

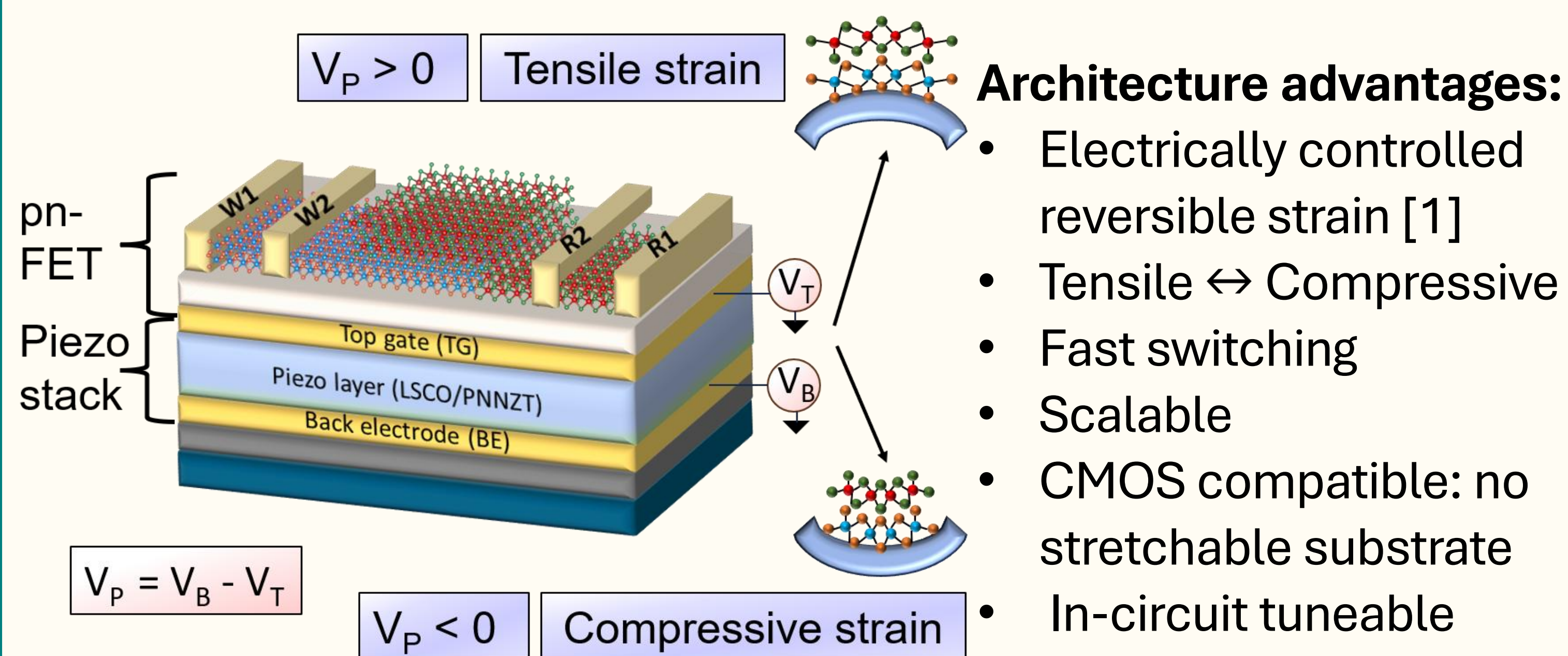
Why WSe₂/ReS₂ heterostructure?

- Ambient stable
- CMOS/MEMS fabrication compatible
- High optical absorption in Vis-NIR band
- WSe₂ with Pt contact: ambipolar
- ReS₂ with Pt contact: n-type transport
- Forms Type-II heterojunction



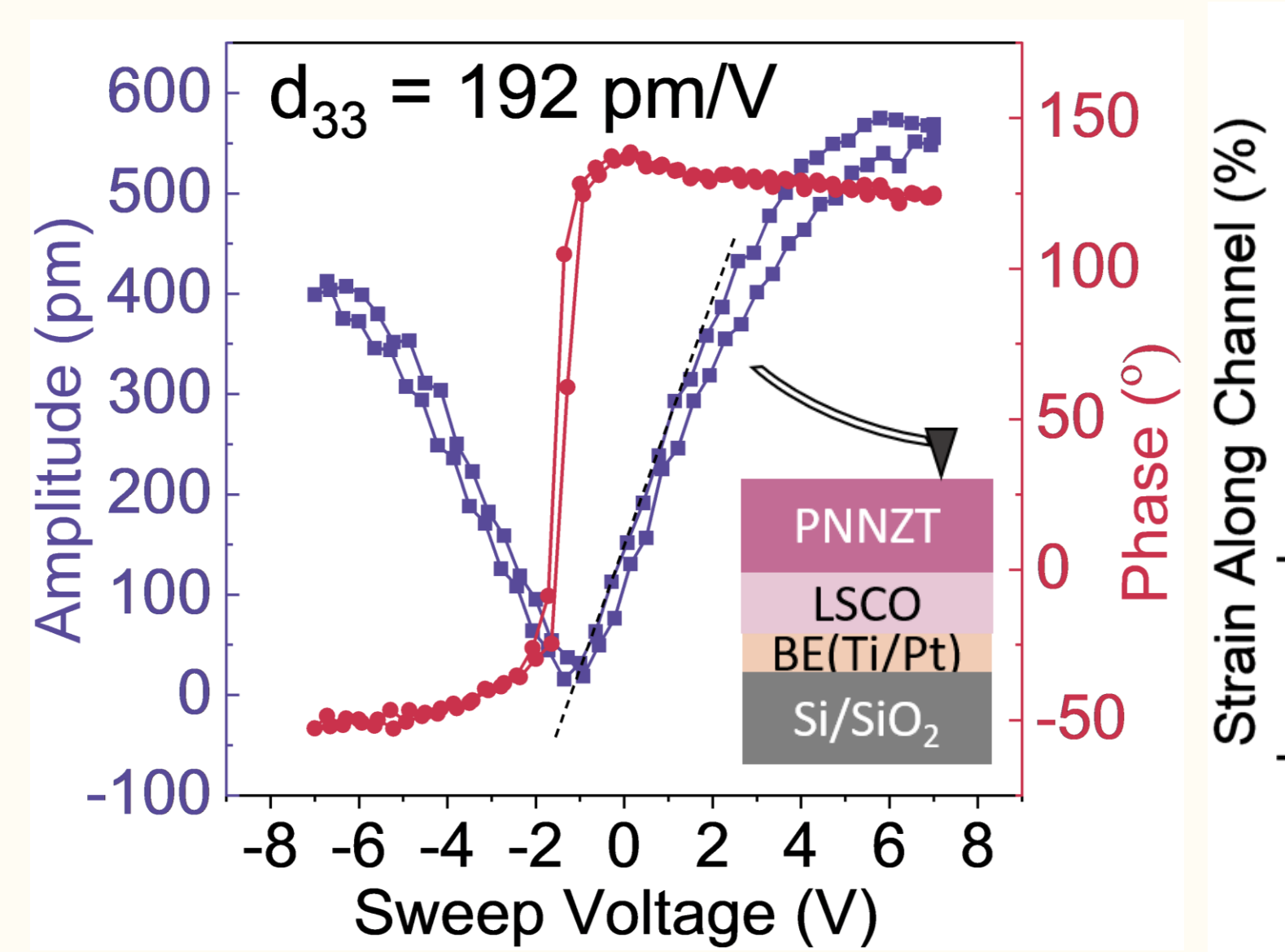
Device Architecture

Converse Piezoelectric Effect: Electrical bias → Material deformation

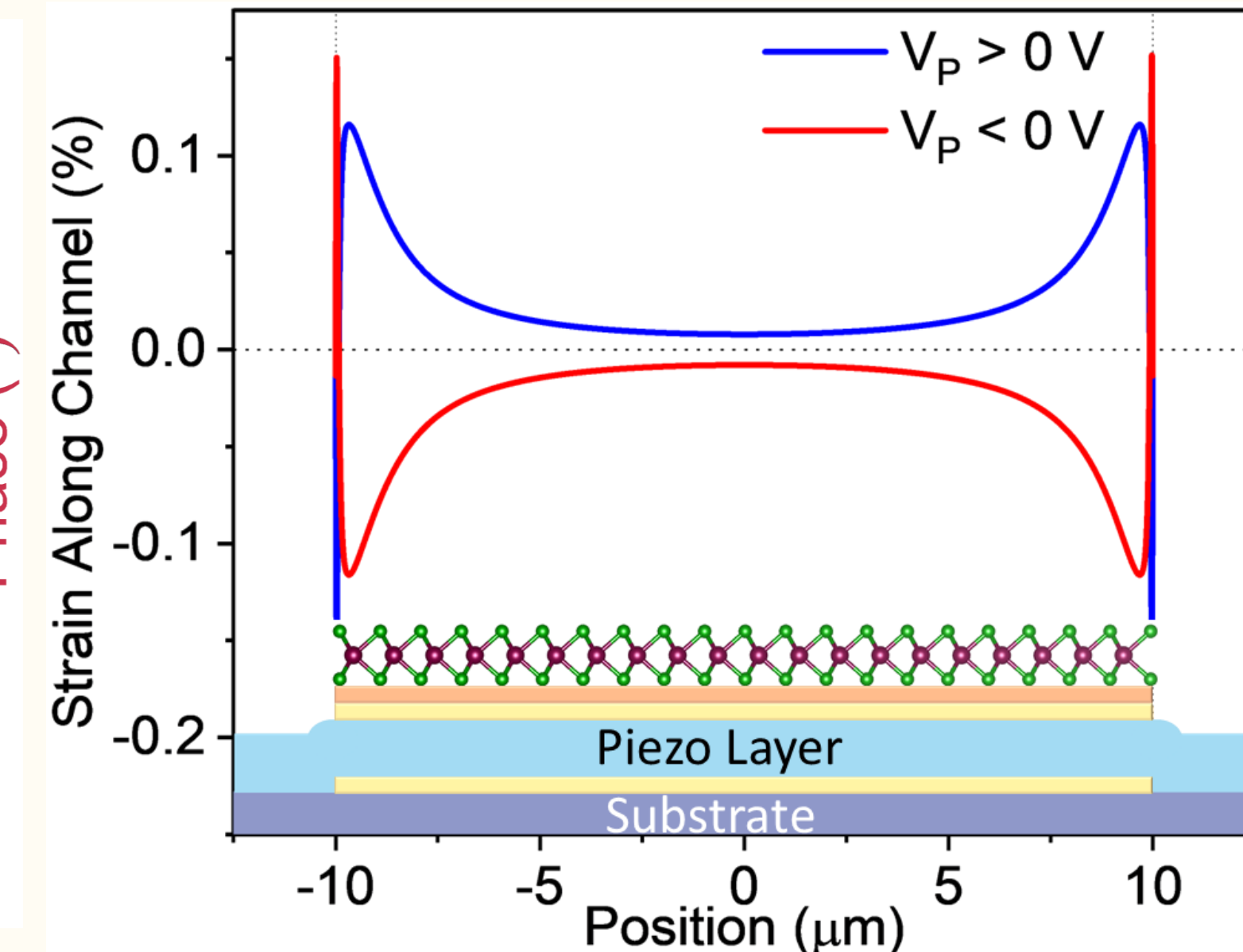


Strain Characterization and Strain Transfer

PFM characterization



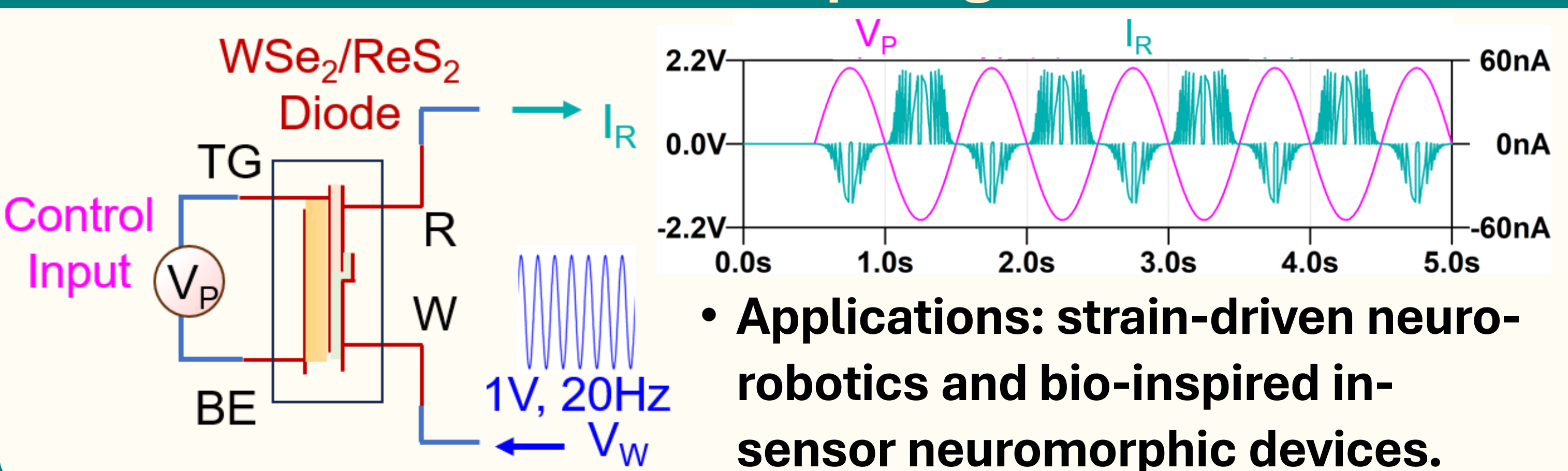
Strain transfer FEM simulation



- PFM shows a high piezoelectric coefficient in the z-direction

- V_P > 0: Tensile Strain
- V_P < 0: Compressive Strain

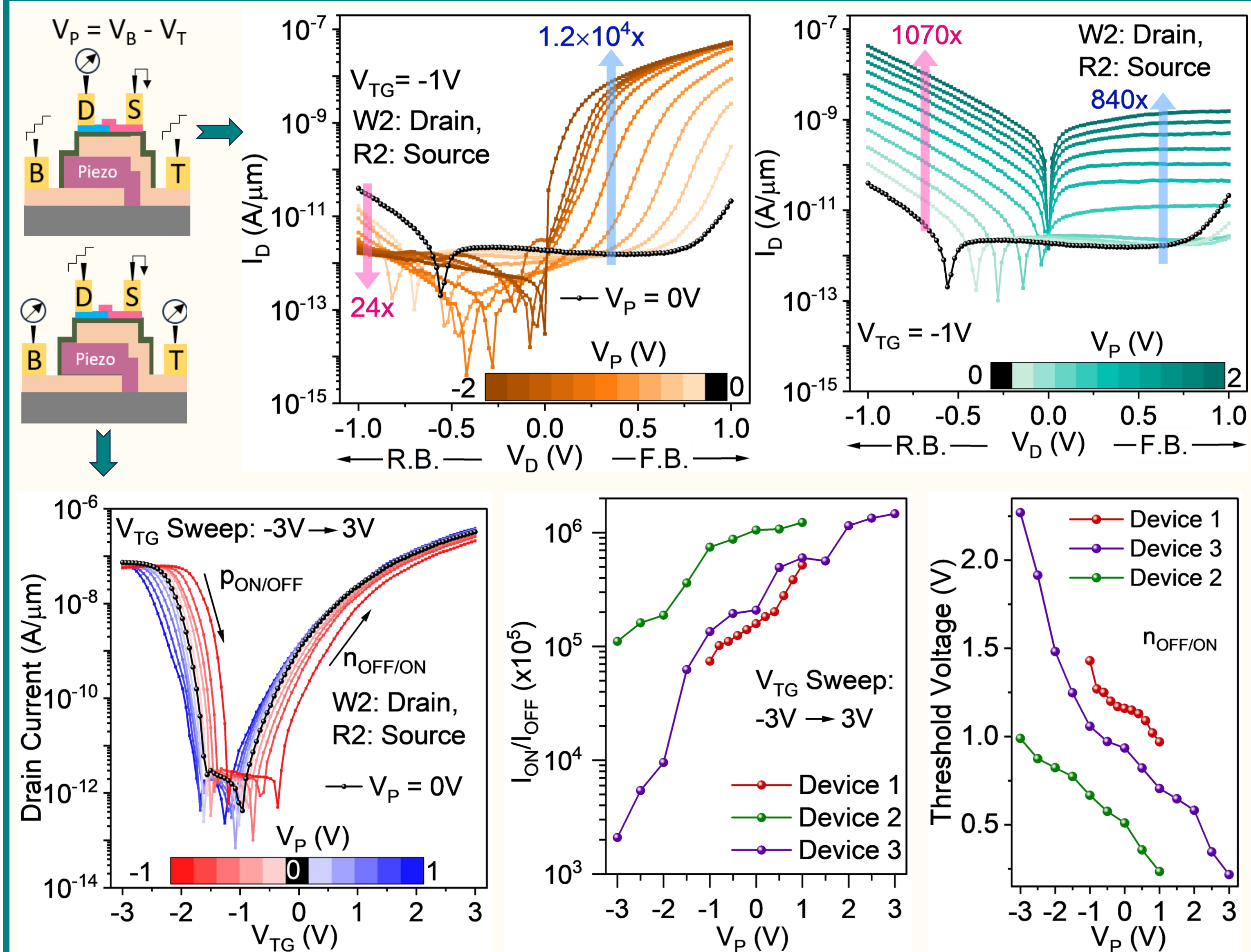
Strain-Induced Spiking Neuron



Performance Benchmarking

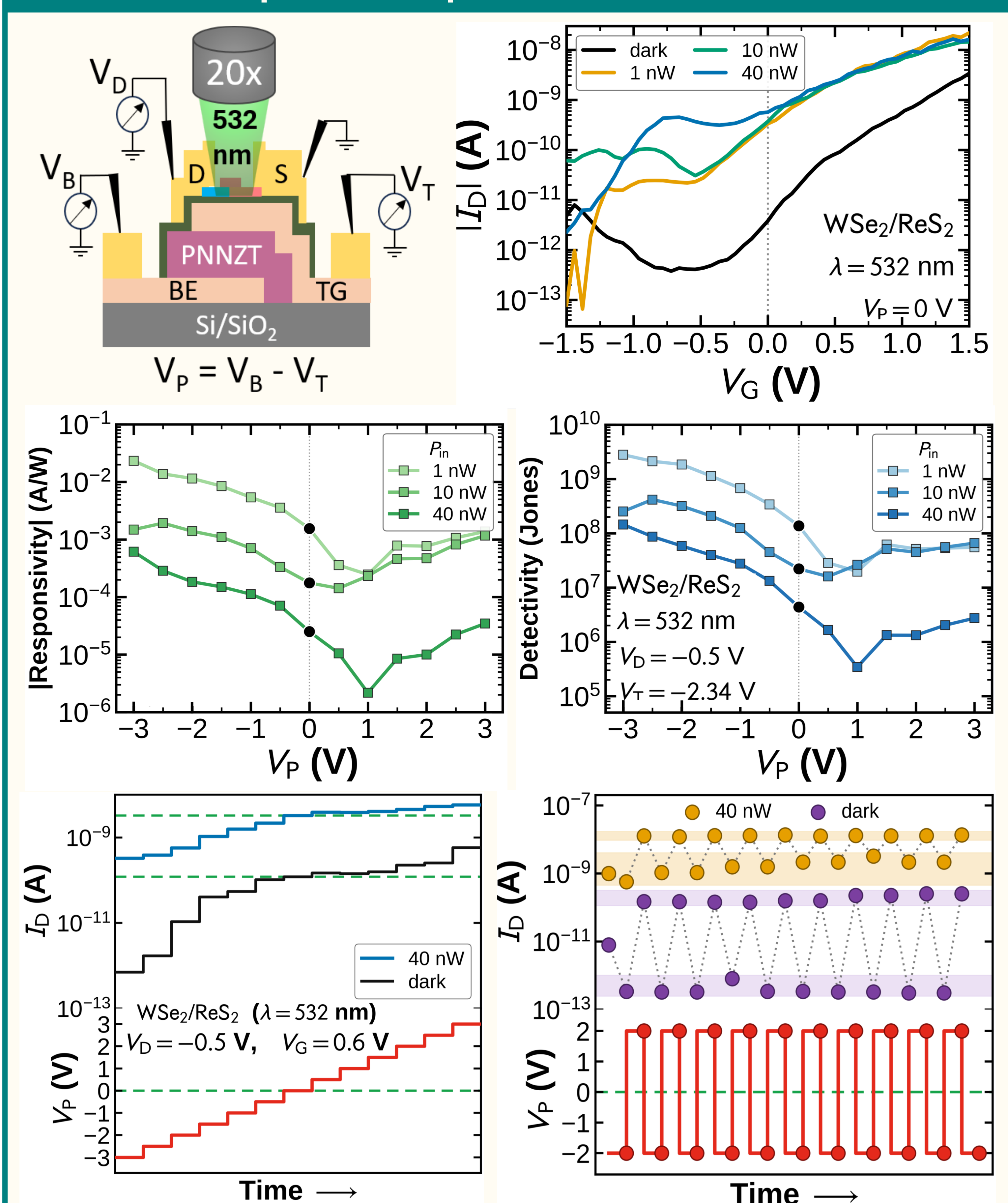
Ref	Substrate-Method	Material	Responsivity change	CMOS fab.
2	PET-Flexible subs.	n-WSe ₂ /WSe ₂	800 nm, 216.7 mA/W, 455%	No
3	PET-Flexible subs.	SnS/MoS ₂	554 nm, 211 mA/W, ~97%	No
4	ZnO-ZnO nano rods	WSe ₂ /Gr	532 nm, ~151 mA/W, ~210%	No
5	Si/SiO ₂ -Epitaxy	WS ₂ /MoS ₂	565 nm, 1850 mA/W, ~200%	No
This	Si/SiO ₂ -Piezo film	WSe ₂ /ReS ₂	532 nm, 37 mA/W, 9400%	Yes

Strain-dependent Hetero-transistor Performance



- Direction of pn-FET I_DV_D rectification changes with V_P
- Strain modulates essential FET parameters

Strain-dependent Optoelectronic Performance



- Applications: in-sensor computation, I_{ph} amplification and attenuation, artificial retina, and robotic vision.

References and Acknowledgements

References:

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shreyasidas17@gmail.com

